

Senthil Nagappan

AI Experience Design & Human Systems Integration — 18+ years across healthcare, federal, financial services, and enterprise AI.

Frisco, TX (DFW) · remote / open to relocation · 571-403-0835 · arnsk3@gmail.com · www.senthilnagappan.com

Professional Summary

Design leader and front-end practitioner who pairs AI governance, accessibility, and human systems integration to ship safe, human-centered AI. I build the strategic logic, controls, and interfaces that let empathy govern intelligent, non-deterministic systems — then prototype and deliver them in real code. Experience spans clinical AI, federal programs, financial-services model risk, and enterprise design systems serving millions.

Core Competencies

AI governance & model risk · NIST AI RMF · EU AI Act · ISO/IEC 42001 · SR 11-7 discipline · Human–AI interaction design · Explainability & trust UX · Agentic AI guardrails · Human-in-the-loop oversight · Accessibility leadership (WCAG 2.2 AA/AAA, Section 508) · Design systems · React/TypeScript prototyping · GRC tooling: ServiceNow GRC / Archer, Collibra / Alation, model registries · Federal HSI / DoDI 5000.95 · MIL-STD-1472H · IEC 62366

Experience

AI Experience Designer & HSI Lead — GE HealthCare (Brain Pulse Inc)

Mar 2025 – Present · Frisco, TX (remote)

Agentic clinical workflows, human–AI trust and explainability, design-to-code prototyping in React/TypeScript, and design-system components adopted across 10+ teams.

Sr. UX & Accessibility Lead — Social Security Administration (Leidos)

Oct 2022 – Feb 2025 · Remote

Enterprise design system across 15+ modules serving 50M+ users; traced 65% of interaction issues to 8 reused components, saving \$1.5M+ annually; WCAG 2.2 AA / Section 508 across 40+ monthly releases.

Sr. UX & Inclusive Design Lead — Best Buy Health

Sep 2021 – Sep 2022 · Minneapolis, MN (remote)

Connected medical-device experiences for 1M+ users, summative validation feeding 510(k) human-factors evidence, BRIX design system.

Sr. UX & Program Lead — HHS / SAMHSA (Eagle Technologies)

Mar 2014 – May 2021 · Remote

Behavioral-health data platforms and visualization for 2M+ users; led cross-functional delivery of federal digital services.

Earlier Roles

AAMC · CareFirst BlueCross BlueShield · JPMorgan Chase · Morgan Stanley — UX, accessibility, and design-system leadership.

Selected AI & Prototyping Work

Full-stack AI visual-assistance PWA for blind and low-vision users (Vite + React with serverless AI proxy) featuring real-time scene description, OCR reading, and product identification; enterprise AI-experience pattern library (explainability, trust calibration, agentic hand-off, human-in-the-loop); interactive design-system and data-visualization prototypes.

Education & Certifications

B.E., Electronics & Communication Engineering (HCI / human-factors focus), Madurai Kamaraj University, 1999.

AIGP — AI Governance Professional (IAPP) (in progress) · IAAP CPACC + WAS (CPWA) · CUA · CSM · Salesforce Certified UX Designer (in progress)

Standards & Frameworks

WCAG 2.2 AA/AAA · Section 508 · ADA · FDA HFE / IEC 62366 · ISO 14971 · MIL-STD-1472H · MBSE/SysML · DoDI 5000.95 · NIST AI RMF · EU AI Act · ISO/IEC 42001 · SR 11-7

Leadership & Community

Vice President, Texas Astronomical Society (500+ member nonprofit); Cubmaster & Assistant Scoutmaster, Scouting America (Wood Badge trained, Spark Plug Award 2025); Lead Coach, FIRST LEGO League.